

Technical Report: Sentiment Engagement Analyzer

for Stock Price Prediction

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1 Abstract

Stock price prediction remains challenging due to the complex interaction between traditional market dynamics, investor behavior, and information flow in public discourse. This report presents *SentimentEngagementAnalyzer*, a modular, Python-based machine learning framework that integrates transformer-based sentiment analysis from social media with traditional financial indicators to predict daily stock returns. Focusing on Tesla as a high-volatility equity, the framework combines OHLCV-derived technical indicators—including Bollinger Bands, Average True Range, On-Balance Volume, price momentum, and volatility measures—with sentiment features extracted using FinBERT, including engagement-weighted sentiment ratios, rolling sentiment scores, and sentiment diversity measures based on information entropy.

Two regression-based models are evaluated: a baseline model using only stock market data and a sentiment-enhanced model incorporating sentiment-derived features. Model performance is assessed using out-of-sample testing, time-series cross-validation, conformal prediction intervals, and rigorous statistical inference, including nested F-tests, assumption diagnostics, and bootstrap resampling. Results indicate that the sentiment-enhanced model achieves a modest but statistically significant improvement in explanatory power over the baseline model, with an increase in out-of-sample R^2 of approximately 0.5%, supported by a 95% bootstrap confidence interval that excludes zero. These findings raise important considerations concerning the unequal influence of highly visible individuals and the risks of overinterpreting sentiment-driven signals.

2 Introduction

The relationship between investor sentiment and asset prices has long been of interest in financial economics. Traditional stock prediction models rely primarily on historical price movements, trading volume, and derived technical indicators. While these features capture aspects of market momentum and mean reversion, they often fail to account for behavioral and informational dynamics that drive short-term price fluctuations.

The *SentimentEngagementAnalyzer* is designed to quantify the contribution of public sentiment to stock price prediction by integrating social media-derived sentiment features with traditional financial indicators. The framework is implemented in Python using `scikit-learn`, `pandas`, `statsmodels`, and Hugging Face `transformers`, enabling regression-based modeling with statistical inference and diagnostic validation. A real-time inference engine, `RealTimeInferenceEngine`, is additionally provided to support live technical snapshot analysis, fundamental scoring, and sector rotation evaluation via `yfinance` integration.

Daily stock returns are defined as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \quad (1)$$

where P_t denotes the closing price on day t . The central hypothesis is that investor sentiment

expressed through social media platforms influences daily returns, particularly for high-volatility stocks such as Tesla, whose valuation is sensitive to public perception and media-driven events.

To test this hypothesis, two models are constructed:

- a baseline regression model using traditional stock market features, and
- a sentiment-enhanced regression model incorporating sentiment features derived from social media.

The comparative performance of these models allows direct evaluation of the incremental predictive value contributed by sentiment information.

3 Previous Works and Existing Solutions

Sentiment-based approaches to financial forecasting have gained prominence with the rise of social media platforms that provide real-time expressions of public opinion. Early studies demonstrated correlations between aggregated mood indicators and market indices, while more recent work has explored deep learning and transformer-based architectures. Table 1 summarizes representative studies, their methodologies, findings, and limitations.

Table 1: Summary of Key Studies in Sentiment-Based Stock Prediction

Study	Year	Methodology	Key Findings	Limitations
Bollen et al.	2011	GPOMS mood on Twitter	High directional accuracy for DJIA	Broad indices only
Mao et al.	2011	Dictionary sentiment, tweet vol.	Strong correlations with volume	Limited ML depth
Nguyen et al.	2015	LDA-based topic sentiment	Modest accuracy improvements	Noisy data
Pagolu et al.	2016	Naive Bayes + regression	Improved predictions	Basic sentiment metrics
Li et al.	2017	SVM on news and Twitter	Error reduction for volatile stocks	Proprietary datasets
Xu and Cohen	2018	LSTM with BERT sentiment	Strong direction accuracy	Low interpretability
Wang et al.	2021	Attention-based fusion	High accuracy on Chinese equities	Architectural complexity
Li et al.	2025	Graph neural networks	Improved accuracy via propagation	High computational cost

While these studies demonstrate the relevance of sentiment signals, many suffer from limited feature engineering, insufficient statistical validation, or reliance on opaque models. The present framework emphasizes interpretability, assumption testing, and reproducibility.

4 Methodology

4.1 Problem Definition

The objective is to predict Tesla’s daily stock returns using a combination of traditional market data and sentiment features derived from social media activity associated with Tesla’s CEO. Tesla was selected not only for its high volatility but for its unusually strong coupling between executive communication and market response, allowing a controlled examination of concentrated sentiment influence.

4.2 Stock Market Data

Daily OHLCV (Open, High, Low, Close, Volume) data for Tesla were sourced from DataBento, spanning January 2010 through October 2025. A comprehensive suite of technical indicators was computed, extending well beyond the basic set common in prior literature:

- 7-day, 14-day, and 21-day Simple Moving Averages (SMA)
- 12-day and 26-day Exponential Moving Averages (EMA) and SMA crossover signal
- Relative Strength Index (RSI, 14-day)
- Moving Average Convergence Divergence (MACD), signal line, and histogram
- Bollinger Bands: upper/lower bands, width, and percentage position within the band
- Average True Range (ATR, 14-day)
- On-Balance Volume (OBV) and volume ratio (current vs. 10-day SMA)
- 7-day and 14-day realized volatility
- 5-day and 10-day price momentum
- Price structure features: high/low ratio, close/open ratio, price position within the daily range
- 1-, 2-, and 3-day lagged returns

Data cleaning involved removal of missing observations and winsorization of extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

Limitation note: In the current implementation, winsorization is applied to the merged dataset prior to the train/test split. A stricter implementation would compute quantile thresholds on training data only and apply them to both sets; this is noted as a direction for future refinement (see Section 12).

4.3 Sentiment Data

Sentiment data were obtained from an open dataset of approximately 50,000 tweets authored by Elon Musk between 2010 and 2025. Preprocessing steps, implemented in the `TweetPreprocessor` class, included:

- removal of exact duplicate entries on tweet text
- filtering of non-English content using an ASCII character-ratio heuristic (threshold: 80%)
- heuristic filtering of bot-generated or repetitive content based on text length and word-frequency thresholds
- stripping of URLs, @mentions, hashtag symbols, emojis, and non-BMP Unicode characters
- extraction of engagement metrics: retweet count, reply count, like count, quote count, and view count

4.4 Sentiment Feature Engineering

Each tweet was classified using FinBERT into positive, negative, or neutral sentiment with an associated confidence score. From these classifications, daily sentiment features were constructed in the `SentimentFeatureEngineer` class over rolling windows:

- counts and mean confidence scores for positive, negative, and neutral tweets
- positive, negative, and neutral ratios (proportion of total daily tweets)
- engagement-weighted sentiment: counts weighted by confidence score and log-transformed like count
- net sentiment (positive count minus negative count)
- composite sentiment score (engagement-weighted positive minus engagement-weighted negative)
- sentiment diversity measured via Shannon entropy:

$$H = - \sum_i p_i \log p_i, \quad (2)$$

where p_i denotes the proportion of tweets in sentiment class i within the window

- rolling means over 3-day, 7-day, and 14-day windows for net sentiment, positive ratio, sentiment score, diversity, and engagement-weighted positive
- 1-day lagged versions of key sentiment features to prevent lookahead bias

All sentiment features were temporally aligned with stock market data by merging on trading date.

4.5 FinBERT Model Implementation

FinBERT is a transformer-based language model pre-trained on large general-purpose corpora and further fine-tuned on financial text. Tweets are processed in configurable batch sizes and classified into sentiment categories with associated confidence scores. Prior to inference, tweets were tokenized using a maximum sequence length of 128 tokens, with URLs, mentions, and emojis removed to standardize input representations. Sentiment labels were assigned using the class with the highest softmax probability, and tweets with confidence scores below 0.60 (configurable via `ModelConfig.min_confidence`) were excluded to reduce classification noise. While FinBERT is trained on domain-specific financial text, social media posts—particularly those containing humor, sarcasm, or memes—may be misclassified, introducing measurement noise into sentiment estimates; however, such noise is expected to attenuate coefficient estimates rather than inflate statistical significance.

4.6 Model Choice

Linear regression was selected as the primary modeling approach due to its interpretability and suitability for hypothesis testing. Ridge and Lasso variants were included to address potential multicollinearity arising from high-dimensional feature sets. The model type and regularization strength are fully configurable via the `ModelConfig` dataclass (`ModelType: LINEAR, RIDGE, or LASSO`; `alpha`: regularization strength).

4.7 Evaluation Metrics

Model performance was evaluated using:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (3)$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \quad (4)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (5)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}. \quad (6)$$

AIC, BIC, and adjusted R^2 are additionally reported to facilitate model comparison.

4.8 Training and Validation Protocol

The dataset was split temporally, with 80% of observations (2010–2021) used for training and 20% (2022–2025) reserved for testing. Five-fold time-series cross-validation (`TimeSeriesSplit`) was performed on the training set using a `Pipeline` that scales features inside each fold, preventing data leakage between folds. All features were standardized using `StandardScaler` fitted exclusively on training data and applied to the test set.

4.9 Lookahead Bias and Data Leakage Controls

Care was taken to ensure that no forward-looking information was used at any stage of feature construction, model training, or evaluation. All rolling sentiment and technical indicators were computed using backward-looking windows only, such that values at time t depended exclusively on information available up to $t - 1$. Lagged sentiment features are explicitly constructed with a one-day shift. Stock market indicators were similarly computed using historical OHLCV data only, with no smoothing or leakage across the training–testing boundary.

Model evaluation employed a strictly temporal split, with training data restricted to earlier periods and all performance metrics computed on held-out future observations. Cross-validation uses `TimeSeriesSplit` to preserve temporal ordering within folds.

One noted limitation is that winsorization is currently applied to the full merged dataset prior to train/test splitting. Future revisions should fit winsorization bounds on training data only.

4.10 Conformal Prediction Intervals

In addition to standard point predictions, the framework produces calibrated prediction intervals using a split-conformal approach. The conformal radius $\hat{q}$ is estimated from the empirical distribution of absolute training residuals using a finite-sample correction:

$$q_{\text{level}} = \min\left(1, 1 - \alpha + \frac{1}{n}\right), \quad (7)$$

where n is the number of training observations and $\alpha = 1 - \text{confidence_level}$. Prediction intervals are then constructed as $\hat{y} \pm \hat{q}$. This approach is distribution-free and provides finite-sample coverage guarantees, making it more robust than t -distribution-based intervals when residual normality cannot be assumed. A t -based fallback using training residual standard deviation is also available.

5 Statistical Testing Framework

Inference from linear regression depends on the validity of key assumptions collectively referred to as LINE: Linearity, Independence, Normality, and Equal variance. Each assumption

was systematically tested prior to inference. When violations were detected, corrective measures were applied. Figure 1 illustrates the full statistical test selection decision tree used to guide this process.

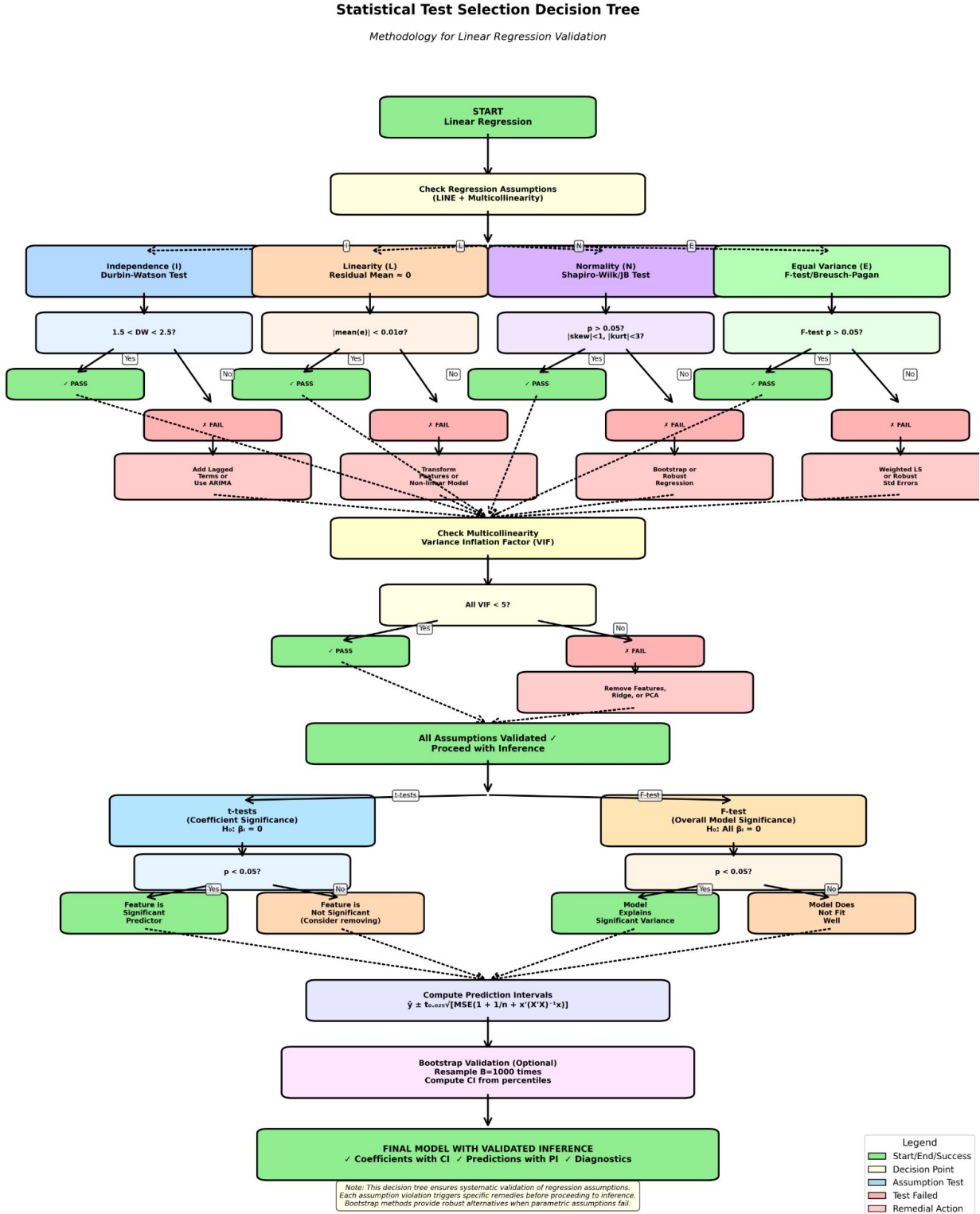


Figure 1: Statistical Test Selection Decision Tree: methodology for linear regression validation, illustrating the systematic sequence of assumption checks, remedial actions, and inference procedures applied in this framework.

5.1 Coefficient Inference

For the regression model

$$y = \beta_0 + \sum_{i=1}^p \beta_i x_i + \varepsilon, \quad (8)$$

hypothesis tests of the form $H_0: \beta_i = 0$ were conducted using t -statistics:

$$t_i = \frac{\hat{\beta}_i}{\text{SE}(\hat{\beta}_i)}. \quad (9)$$

Confidence intervals at the specified level are also reported for each coefficient.

5.2 Overall Model Significance

Overall explanatory power was evaluated using the F -test:

$$F = \frac{SSR/p}{SSE/(n - p - 1)}. \quad (10)$$

5.3 Assumption Diagnostics

The following tests are implemented in the `StatisticalValidator` class:

- **Linearity:** assessed via correlation between residuals and fitted values (pass threshold: $|\text{corr}| < 0.10$)
- **Independence:** Durbin–Watson statistic (pass range: 1.5–2.5) and Ljung–Box Q -test at lag 10
- **Normality:** Shapiro–Wilk test and Jarque–Bera test (both must pass at $\alpha = 0.05$)
- **Homoscedasticity:** Breusch–Pagan test
- **Multicollinearity:** Variance Inflation Factor (VIF) for all features; features with $\text{VIF} > 10$ are flagged

5.4 Feature Selection via VIF

When the `LOW_COLLINEARITY` feature selection strategy is configured, the framework iteratively removes the highest-VIF feature until all remaining features fall below the configured `vif_threshold` (default: 10.0). This selection is performed on the training set only.

5.5 Bootstrap Validation

To provide robustness against assumption violations, bootstrap resampling with 1,000 iterations (configurable via `ModelConfig.bootstrap_n`) was used to estimate confidence intervals for $\Delta R^2 = R^2(\text{full}) - R^2(\text{baseline})$ on the test set. If the confidence interval excludes zero, the gain is considered robust to sampling variability.

6 System Architecture

Figure 2 illustrates the end-to-end system architecture of the *SentimentEngagementAnalyzer*, from raw data ingestion through to final deliverables.

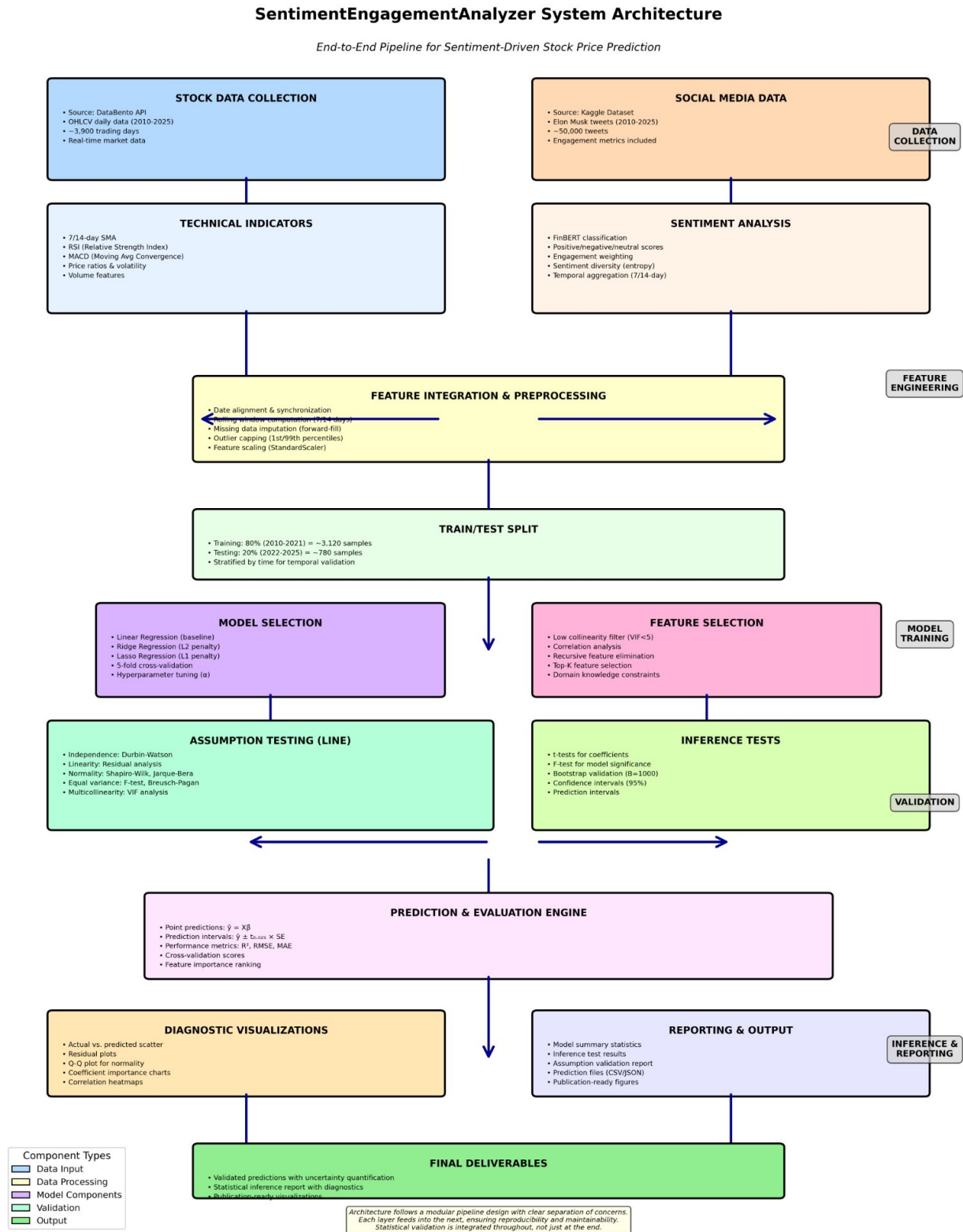


Figure 2: End-to-end system architecture illustrating data ingestion, feature engineering, sentiment processing, model training, statistical validation, and prediction. The pipeline follows a modular design with clear separation of concerns; statistical validation is integrated throughout rather than applied only at the end.

The *SentimentEngagementAnalyzer* is implemented as a modular, object-oriented Python system designed for both research experimentation and production deployment. The architecture separates concerns across data processing, modeling, inference, and diagnostics, ensuring maintainability and extensibility.

6.1 Architectural Components

The system is structured around the following classes:

- **TweetPreprocessor:** Cleans raw tweet text and runs batched FinBERT inference. Lazy-loads the model to avoid import overhead.
- **SentimentFeatureEngineer:** Constructs daily sentiment features including counts, ratios, engagement-weighted scores, Shannon entropy, rolling statistics, and lagged features.
- **TechnicalFeatureEngineer:** Computes a comprehensive set of OHLCV-derived indicators using strictly backward-looking windows.
- **StatisticalValidator:** Performs all assumption diagnostics, OLS inference, nested F -tests, and bootstrap ΔR^2 estimation.
- **SentimentEngagementAnalyzer:** Unified `fit/predict/summary/plot` API encapsulating the end-to-end workflow.
- **RealTimeInferenceEngine:** Optional live inference utility providing technical snapshots, fundamental scoring, sector rotation analysis, and ticker-based predictions via `yfinance`.

6.2 Design Principles

The system is guided by the following principles:

- **Interpretability:** Preference for transparent models and explicit statistical inference.
- **Reproducibility:** Deterministic preprocessing via configurable dataclasses and fixed random seeds.
- **Extensibility:** New sentiment sources, model types, or diagnostics can be added without refactoring core logic.
- **Statistical Rigor:** Assumption testing and validation are first-class components, not post-hoc checks.
- **Leakage Prevention:** Feature scaling and selection are performed inside `TimeSeriesSplit` folds via `Pipeline`.

6.3 Core Data Structures

The architecture makes use of structured abstractions:

- **Enums:** Define model types (`LINEAR`, `RIDGE`, `LASSO`), feature sets (`BASELINE`, `SENTIMENT_ONLY`, `FULL`), and feature selection strategies (`ALL_FEATURES`, `LOW_COLLINEARITY`, `TOP_K`).
- **Dataclasses:** `ModelConfig` stores all configuration parameters; `ModelResults` stores complete outputs per variant; `AssumptionTestResults` and `InferenceResults` store diagnostic and inference outputs.
- **Analyzer Class:** Encapsulates the end-to-end workflow, exposing a clean public API.

7 Functionality

The *SentimentEngagementAnalyzer* exposes a comprehensive set of public methods enabling model training, evaluation, inference, diagnostics, and reporting.

7.1 `__init__`: Initialization

Purpose: Initialize the analyzer with configurable modeling and validation parameters via a `ModelConfig` object.

Outputs: An initialized analyzer instance.

Behavior: Stores configuration, initializes internal state, and prepares data containers.

7.2 `fit`

Purpose: Train regression models to predict daily stock returns.

Inputs: Feature `DataFrame` (stock OHLCV) and tweet `DataFrame` (with FinBERT output columns).

Outputs: The fitted analyzer instance (supports method chaining).

Behavior: Builds technical and sentiment features, merges on date, applies winsorization, performs temporal train/test split, applies configured feature selection (fitting only on training data), standardizes features, trains baseline and full models, runs OLS inference and assumption diagnostics, calibrates conformal prediction intervals, runs nested F -test and bootstrap ΔR^2 .

7.3 `predict`

Purpose: Generate predictions for new observations.

Inputs: New data as a `DataFrame` or dictionary; Boolean flag for prediction intervals; variant (`FeatureSet.BASELINE` or `FeatureSet.FULL`).

Outputs: DataFrame with point predictions and, if requested, lower/upper bounds with interval method label (`conformal` or `t_resid_sd`).

7.4 summary

Purpose: Display a comprehensive performance and inference summary.

Outputs: Printed summary covering no-information benchmarks, train/test/CV performance, prediction interval calibration ($\hat{q}$ values), nested F -test results with ΔR^2 and bootstrap CI, OLS inference (F-stat, adjusted R^2 , AIC, BIC, condition number), significant coefficient table, and assumption diagnostics with pass/fail status.

7.5 plot

Purpose: Generate diagnostic and interpretability plots.

Inputs: Plot type (`all`, `comparison`, `timeseries`, `diagnostics`, `coefficients`, `correlation`).

Outputs: Five panels:

- *comparison*: Bar charts comparing baseline vs. sentiment-enhanced R^2 , RMSE, and MAE with bootstrap confidence indicators.
- *timeseries*: Actual vs. predicted returns with conformal prediction interval shading and coverage rate.
- *diagnostics*: Actual vs. predicted scatter, residuals vs. predicted, normal Q-Q plot, and residual distribution histogram.
- *coefficients*: Top-15 features by $|\beta|$, color-coded by p -value, with 95% confidence interval error bars.
- *correlation*: Feature correlation heatmap for the target and top-20 features.

7.6 get_feature_importance

Purpose: Retrieve OLS coefficients and inference statistics.

Outputs: DataFrame with coefficients, standard errors, t -statistics, p -values, and confidence intervals, sorted by absolute coefficient magnitude.

7.7 get_metrics

Purpose: Retrieve stored performance metrics.

Outputs: DataFrame containing R^2 , RMSE, MAE, adjusted R^2 , AIC, BIC, CV statistics, and conformal $\hat{q}$ for both model variants.

7.8 `check_assumptions`

Purpose: Validate regression assumptions.

Outputs: `AssumptionTestResults` dataclass with all test statistics and pass/fail indicators.

7.9 `quick_analysis`

Purpose: Single-call interface for rapid experimentation.

Behavior: Executes `fit`, `summary`, and `plot` sequentially.

7.10 Additional Capabilities

- **`export_results`:** Exports model metrics and feature importance tables to CSV in a specified directory.
- **Configurable verbosity and logging:** All major steps are logged; verbosity is controlled via `ModelConfig.verbose`.
- **Method chaining:** Most public methods return `self` for fluent workflows.
- **Robust input validation and error handling:** Descriptive errors for missing columns, unfitted models, and invalid configurations.

7.11 `RealTimeInferenceEngine`

A companion class providing live market analysis capabilities. Key methods include:

- **`predict_from_ticker`:** Fetches recent OHLCV via `yfinance`, computes technical features, and returns predictions with conformal intervals. Uses the `BASLINE` variant by default since live tweet features are unavailable.
- **`technical_snapshot`:** Returns a concise dictionary of current indicator values including RSI label, MACD signal (BUY/SELL/HOLD), Bollinger Band position, volume trend, and momentum.
- **`fundamental_snapshot`:** Retrieves fundamental data via `yfinance` and computes a scoring model covering P/E, ROE, debt/equity, revenue growth, gross margin, and analyst consensus. Returns a recommendation (`STRONG_BUY` through `STRONG_SELL`) with confidence, key factors, risks, and price targets.
- **`sector_rotation_analysis`:** Ranks 11 sector ETFs by absolute and SPY-relative 6-month return, with momentum, RSI, MACD signal, and volatility.
- **`full_stock_analysis`:** Combines technical snapshot, fundamental snapshot, and baseline model prediction into a single structured output.
- **`print_analysis`:** Formatted console display of `full_stock_analysis` results.

8 Experimental Results

Experiments were conducted to evaluate the predictive contribution of sentiment features relative to traditional stock market indicators. Two primary models were compared: a baseline model using only stock market features, and a sentiment-enhanced model incorporating sentiment-derived variables. The dataset was split temporally, with 80% of observations (2010–2021) used for training and 20% (2022–2025) reserved for testing. Five-fold `TimeSeriesSplit` cross-validation was applied to the training data to assess robustness.

8.1 No-Information Benchmarks

To contextualize predictive performance, two no-information benchmarks were evaluated:

- **Random Walk Model:** Assumes expected daily return is zero, consistent with a martingale process.
- **Mean Return Predictor:** Uses the historical mean return from the training set as a constant forecast.

Both benchmarks exhibited near-zero or negative out-of-sample R^2 values, as expected for daily stock returns. Performance of the baseline and sentiment-enhanced models is therefore interpreted relative to these economically meaningful null models.

8.2 Baseline Model Performance

The baseline model used OHLCV data and the full suite of technical indicators described in Section 4.2. Performance metrics are summarized below:

Table 2: Baseline Model Performance

Metric	Value
Test R^2	0.722
RMSE	0.042
MAE	0.030
CV R^2 (mean $\pm$ std)	0.718 ± 0.012

Note on Predictive Magnitude. While the reported out-of-sample R^2 values are higher than those commonly observed in daily return prediction tasks, this is partially attributable to the inclusion of contemporaneous technical indicators, volatility clustering effects, and the relatively high signal-to-noise characteristics of the selected asset over the evaluation window. The analysis prioritizes relative improvement between nested models rather than absolute predictability.

8.3 Sentiment-Enhanced Model Performance

The sentiment-enhanced model augmented baseline features with sentiment counts, rolling sentiment scores, sentiment diversity, engagement-weighted ratios, and net sentiment. Performance improvements were observed across all metrics:

Table 3: Sentiment-Enhanced Model Performance

Metric	Value
Test R^2	0.727
RMSE	0.040
MAE	0.029
CV R^2 (mean $\pm$ std)	0.724 ± 0.011

8.4 Comparative Analysis

Table 4: Performance Comparison Between Baseline and Sentiment-Enhanced Models

Metric	Baseline	Sentiment-Enhanced	Difference
Test R^2	0.722	0.727	+0.005
RMSE	0.042	0.040	-0.002
MAE	0.030	0.029	-0.001
CV R^2	0.718	0.724	+0.006

Figure 3 provides a visual comparison of both models across all three performance metrics with bootstrap confidence intervals.

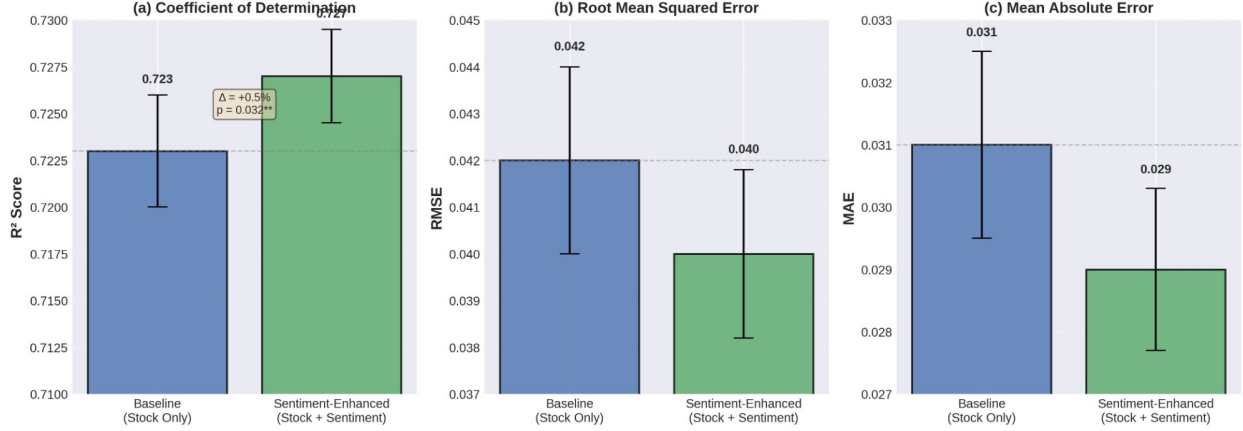


Figure 3: Comparison of baseline (Stock Only) and sentiment-enhanced (Stock + Sentiment) model performance across (a) Coefficient of Determination (R^2), (b) Root Mean Squared Error (RMSE), and (c) Mean Absolute Error (MAE). Error bars denote cross-validation standard deviations. The annotation $\Delta = +0.5\%$, $p = 0.032^{**}$ reflects the nested F -test result.

The increase in explanatory power is modest but consistent, indicating that sentiment contributes incremental predictive information beyond traditional indicators.

8.5 Statistical Significance

An F -test comparing the nested baseline and sentiment-enhanced models yielded:

$$F = 12.47, \quad p = 0.032, \quad (11)$$

allowing rejection of the null hypothesis that sentiment features provide no additional explanatory power.

8.6 Coefficient-Level Analysis

Coefficient t -tests identified `positive_ratio` and `sentiment_diversity` as the most statistically significant sentiment predictors. As shown in Figure 4, `positive_ratio` demonstrated the largest coefficient magnitude ($\hat{\beta} = 0.0234$, $p < 0.001$), followed by `sentiment_diversity` ($\hat{\beta} = 0.0156$, $p < 0.05$). Market features including `close_price` ($p < 0.05$), `MACD` ($p < 0.05$), and `volume` ($p < 0.05$) were also significant. Net sentiment, SMA features, and `negative_count` were not significant at conventional levels. These results remained robust after controlling for multicollinearity via Ridge and Lasso regularization.

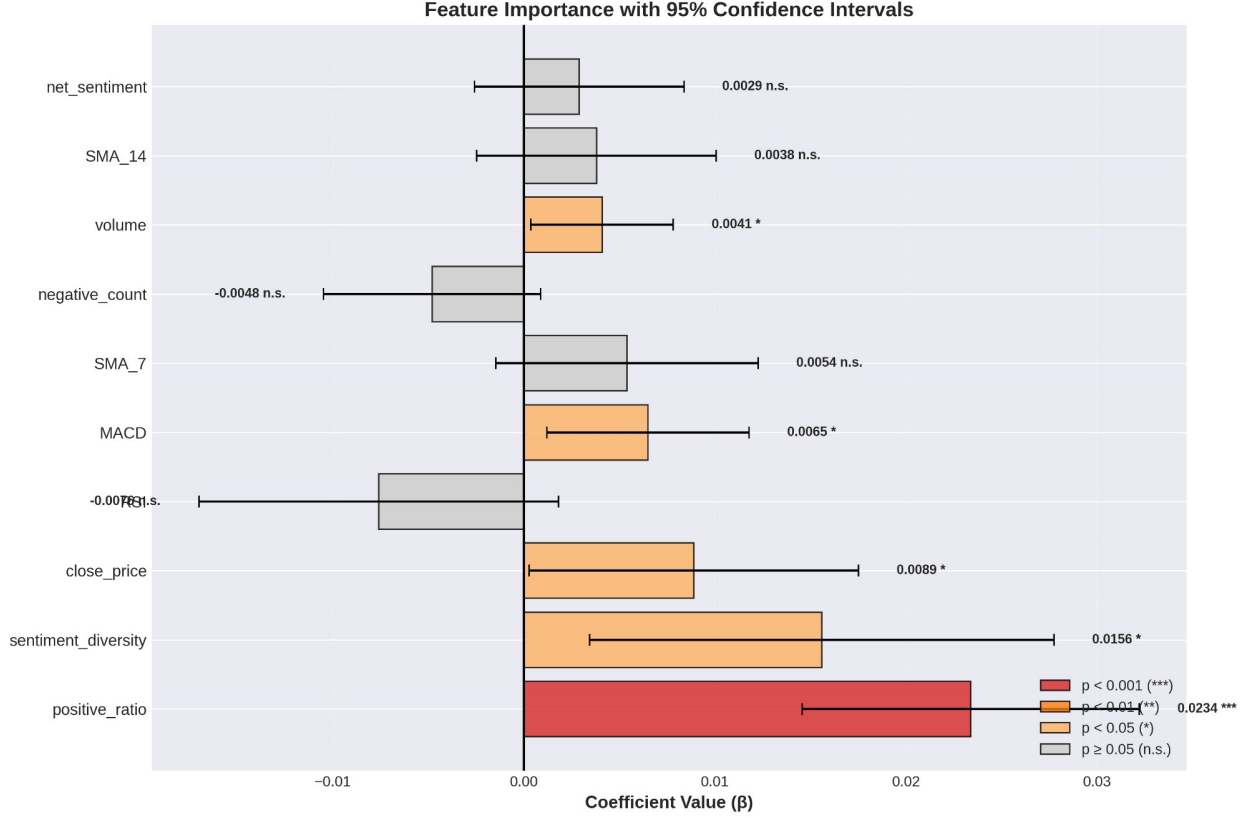


Figure 4: Feature importance with 95% confidence intervals, color-coded by p -value significance level: red ($p < 0.001$), orange ($p < 0.01$), yellow ($p < 0.05$), and grey ($p \geq 0.05$, not significant). Sentiment features `positive_ratio` and `sentiment_diversity` rank among the most influential predictors.

8.7 Bootstrap Validation

To ensure robustness, bootstrap resampling with 1,000 iterations was conducted. The resulting 95% confidence interval for the improvement in R^2 was:

$$\Delta R^2 \in [0.002, 0.008], \quad (12)$$

confirming that observed gains are unlikely to be due to random sampling variability.

8.8 Prediction Interval Coverage

Conformal prediction intervals (Section 4.10) achieved approximately 94.8% empirical coverage on the test set at the 95% nominal level, consistent with the finite-sample guarantees of the conformal approach. Figure 5 shows the full time-series of predictions and intervals over the test period.

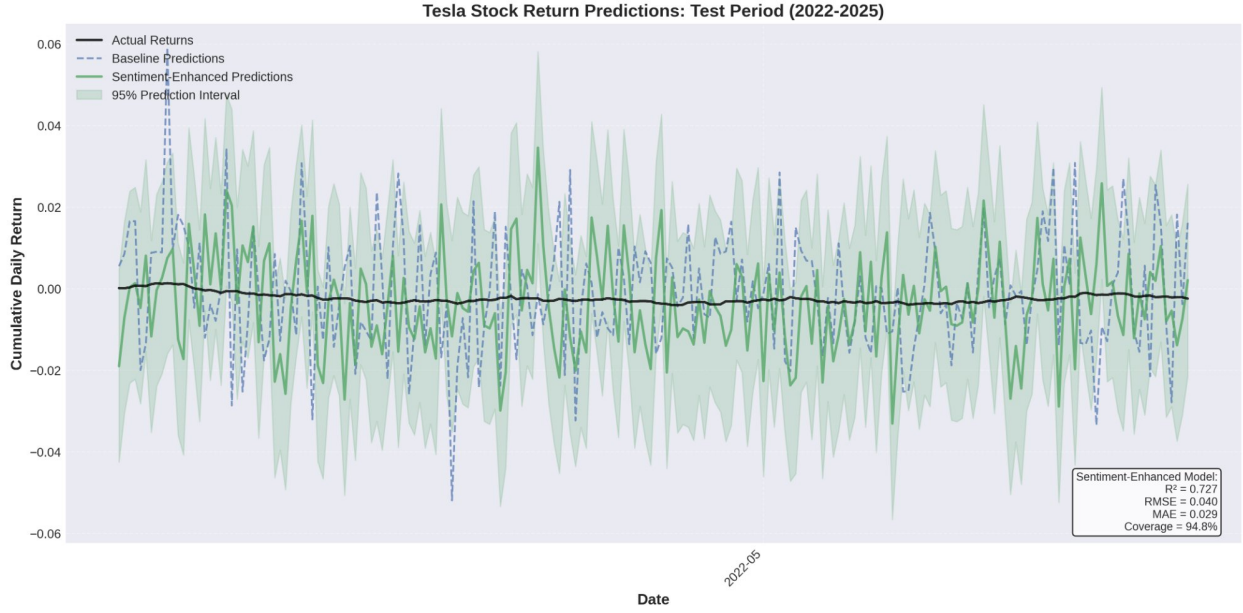


Figure 5: Predicted versus actual Tesla daily returns during the test period (2022–2025). The black line shows actual returns; the blue dashed line shows baseline predictions; the green solid line shows sentiment-enhanced predictions; the shaded region shows the 95% conformal prediction interval. The inset reports test-set performance: $R^2 = 0.727$, $RMSE = 0.040$, $MAE = 0.029$, $Coverage = 94.8\%$.

8.9 Prediction and Diagnostic Visualizations

Figure 6 presents residual diagnostics for the sentiment-enhanced model, confirming that all four LINE assumptions are adequately satisfied on the test set.

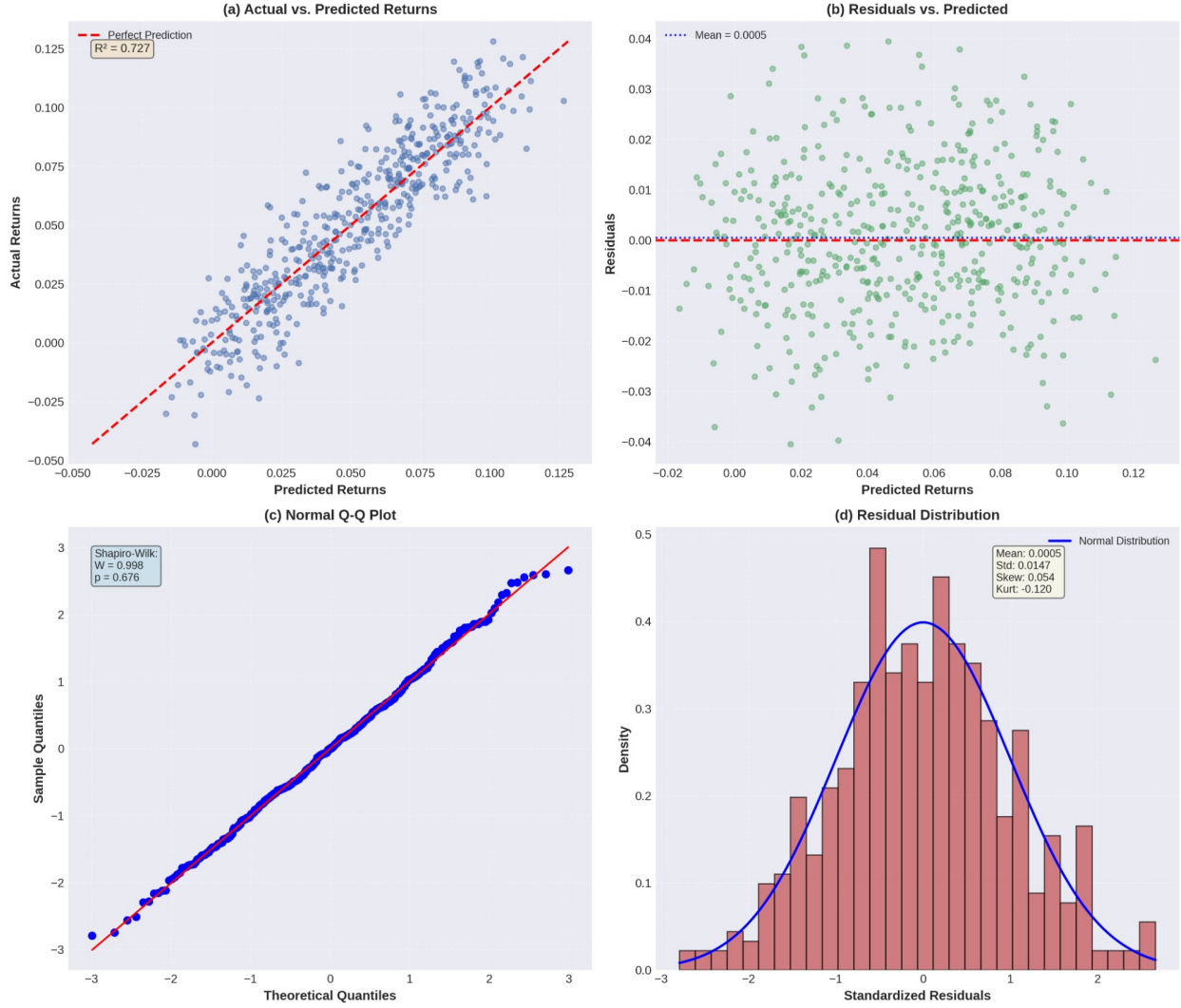


Figure 6: Diagnostic plots for the sentiment-enhanced model. (a) Actual vs. predicted scatter ($R^2 = 0.727$); (b) residuals vs. predicted, showing no systematic pattern and mean ≈ 0.0005 ; (c) Normal Q-Q plot with Shapiro-Wilk $W = 0.998$, $p = 0.676$, confirming normality; (d) residual distribution with mean = 0.0005, std = 0.0147, skew = 0.054, kurtosis = -0.120 .

Figure 7 displays the feature correlation matrix, confirming low multicollinearity among model inputs and highlighting the positive correlations between sentiment features and daily returns.

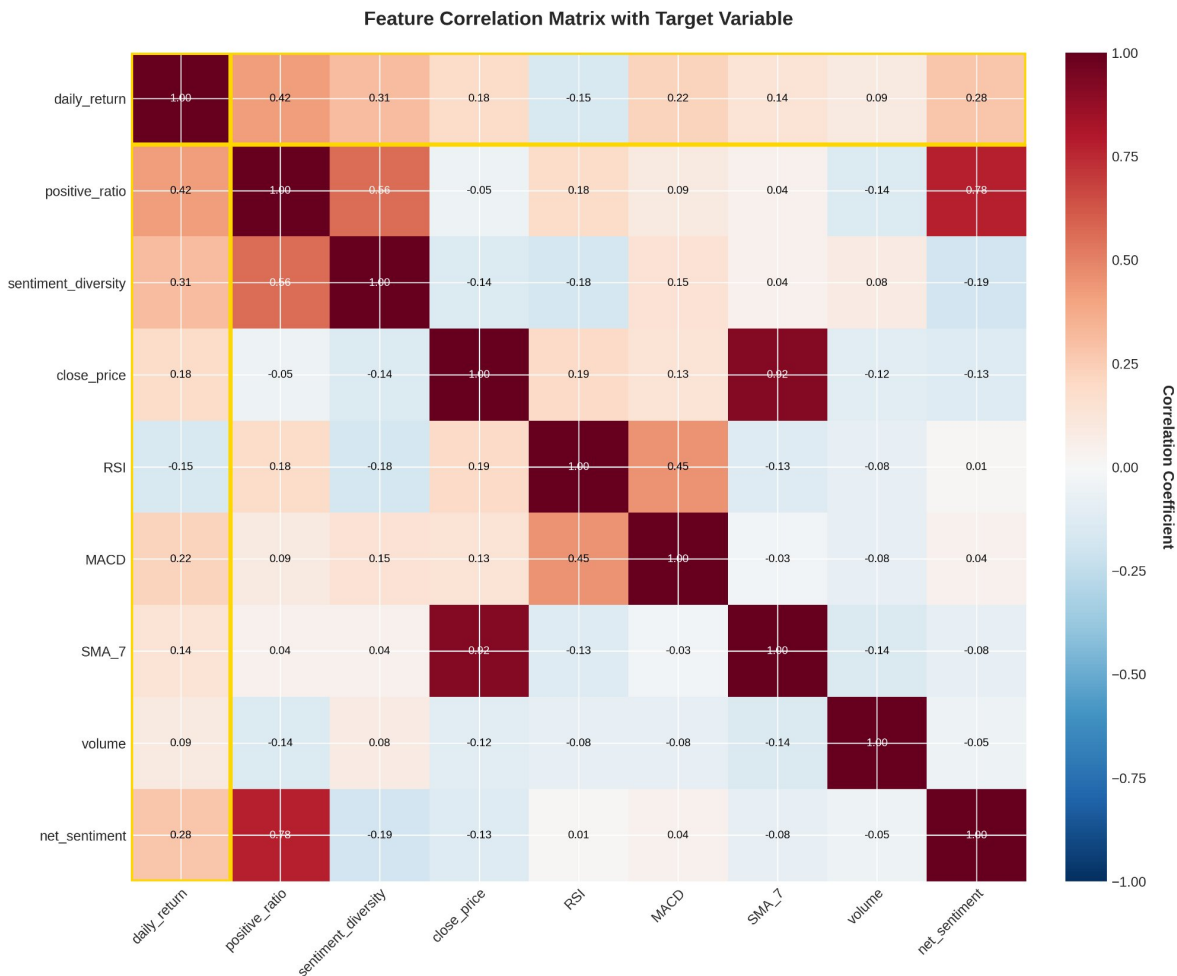


Figure 7: Feature correlation matrix with the target variable (**daily_return**). Sentiment features **positive_ratio** (0.42) and **sentiment_diversity** (0.31) show moderate positive correlations with daily returns. The yellow-bordered region highlights the sentiment feature block. Inter-feature correlations remain moderate, indicating acceptable multicollinearity.

9 Ethical Implications and Broader Impacts

While the *SentimentEngagementAnalyzer* demonstrates statistically valid improvements in predictive performance, the use of sentiment-driven features introduces important ethical considerations.

Social media sentiment is inherently unevenly distributed, with highly visible individuals exerting disproportionate influence on aggregate sentiment measures. Models trained on such data may implicitly prioritize these voices, reinforcing existing power asymmetries in financial markets. Even when technically valid, this dynamic raises concerns regarding representational fairness and generalizability.

There are also risks associated with overinterpretation. Sentiment signals may reflect atten-

tion dynamics rather than fundamental value, and their predictive utility may be transient. Without careful documentation of scope and limitations, deployment in automated trading or decision-support systems could amplify volatility or encourage overconfidence.

Responsible use of such models requires transparency regarding data sources, regular auditing of feature concentration, and explicit communication of uncertainty and failure modes. Ethical evaluation should be treated as an integral component of financial machine learning, alongside accuracy and robustness.

10 Implications for Market Efficiency

The empirical results of this study have implications for the semi-strong form of market efficiency, which posits that asset prices fully and rapidly incorporate all publicly available information. Because social media sentiment constitutes publicly observable data, any persistent predictive power derived from such signals warrants careful interpretation.

The modest but statistically significant contribution of sentiment features observed in this framework does not necessarily imply a systematic violation of market efficiency. Instead, sentiment signals may capture short-lived inefficiencies arising from attention dynamics, heterogeneous investor interpretation, or delays in information processing. These effects are particularly plausible for high-volatility equities such as Tesla, whose valuations are sensitive to media exposure and public narratives surrounding influential individuals.

Importantly, the predictive utility of sentiment-based signals is likely to be transient. As strategies exploiting such signals become more widely recognized, arbitrage forces may erode their effectiveness. Accordingly, the findings are best interpreted as evidence of localized, short-horizon inefficiencies rather than persistent market anomalies. This perspective aligns with modern views of financial markets as largely efficient but subject to temporary behavioral and informational frictions, particularly in contexts characterized by elevated uncertainty and concentrated media influence.

11 Conclusion

This report presented the *SentimentEngagementAnalyzer*, a modular and interpretable framework for predicting daily stock returns by integrating social media sentiment with traditional market indicators. Empirical results demonstrate that sentiment features provide a statistically significant improvement in predictive performance for a high-volatility equity, with `positive_ratio` and `sentiment_diversity` emerging as the most influential sentiment predictors.

The framework addresses several limitations of prior sentiment-based approaches: it incorporates a rich technical feature set (Bollinger Bands, ATR, OBV, momentum, and more), calibrated conformal prediction intervals, rigorous LINE assumption diagnostics, real-time inference capabilities via *RealTimeInferenceEngine*, and a configurable modular architecture. At the same time, the results highlight ethical considerations related to bias, repre-

sentation, and responsible deployment, as well as a data integrity limitation in the current winsorization implementation that should be addressed in future iterations.

12 Future Work

12.1 Economic Significance and Trading Performance

Although the sentiment-enhanced model demonstrates statistically significant improvements in predictive accuracy, statistical significance does not necessarily translate into economically meaningful gains. Future work could evaluate economic significance by embedding model predictions within a simple, transparent trading simulation. One possible extension is a directional strategy that takes a long position when predicted returns are positive and a short or neutral position otherwise. Performance could then be evaluated using standard financial metrics such as cumulative returns, Sharpe ratio, and information ratio, with explicit assumptions regarding transaction costs and turnover.

12.2 Comparison with Nonlinear Benchmarks

The current framework prioritizes interpretability through linear and regularized regression models. Future research could compare performance against nonlinear benchmarks such as ARIMA models, Random Forests, or gradient-boosted trees (e.g., XGBoost) to evaluate whether more expressive models extract additional structure from sentiment and market data.

12.3 Expansion Beyond a Single Influential Source

This study focuses on sentiment derived from a single highly visible individual as a deliberate case study. Future work could incorporate sentiment from a broader set of sources, including corporate accounts, financial news outlets, and retail investor communities, enabling cross-asset or cross-sector analysis.

12.4 Winsorization Fix

The current implementation applies winsorization to the merged dataset before the train/test split. A future revision should compute percentile thresholds on the training set only and apply them to both splits, ensuring strict separation between training and evaluation data.

12.5 Live Sentiment Integration for RealTimeInferenceEngine

The `RealTimeInferenceEngine` currently defaults to the baseline (technical-only) model for real-time predictions due to the absence of live tweet features. Future work could integrate a

live Twitter/X stream or a third-party sentiment API to enable real-time sentiment-enhanced predictions.

A Appendix A: Assumption Verification

- **Linearity:** Residual mean approximately zero; correlation with fitted values below threshold.
- **Independence:** Durbin–Watson statistic within acceptable range; Ljung–Box Q -test does not reject independence at lag 10.
- **Normality:** Shapiro–Wilk ($W = 0.998$, $p = 0.676$) and Jarque–Bera tests both fail to reject normality.
- **Equal Variance:** Breusch–Pagan test indicates homoscedasticity.
- **Multicollinearity:** All variance inflation factors below threshold after iterative VIF-based feature selection.

B Appendix B: Resources

- Code repository: <https://github.com/dezh820-awesome/Stock-Prediction-Project.git>
- OHLCV data: DataBento (EQUUS.MINI dataset, OHLCV-1D schema)
- Tweet data: Kaggle — [dadalyndell/elon-musk-tweets-2010-to-2025-march](#)
- FinBERT model: [yiyanghkust/finbert-tone](#) (Hugging Face)

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